Options - One-Page Reference

Standard academic notation (Hull / Black-Scholes). Payoffs per share; x100 for one contract.

Variables

```
S        underlying price.  S0 = spot now,
         S_T = price at expiration
K        strike   (older texts: X or E)
c, p     European call / put premium
C, P     American call / put premium
T        time to expiry (yrs);  tau = T - t
r        risk-free rate (cont. compounded)
sigma    volatility
q, D     dividend yield / PV of dividends
N(.)     standard-normal CDF
```

Payoff at expiry (profit, per share)

```
Long call    max(S_T - K, 0) - c
Short call   c - max(S_T - K, 0)
Long put     max(K - S_T, 0) - p
Short put    p - max(K - S_T, 0)
```

Pattern: long = intrinsic - premium; short flips the sign. Calls use S_T-K, puts use K-S_T. Rigorous profit compounds the premium: ... - c * e^(rT).

Put-call parity (European, no div.)

`c + K * e^(-rT)  =  p + S0`

Rearranged: c - p = S0 - K*e^(-rT). Long call + short put at same K = a synthetic forward on the stock. With dividends, use S0 - D in place of S0.

The Box Spread (synthetic loan)

Pick K1 < K2, same expiry. Stack two verticals:

- Bull call spread: long call K1, short call K2
- Bear put spread: long put K2, short put K1

Combined payoff for ANY S_T:

```
 max(S_T-K1,0) - max(S_T-K2,0)
  + max(K2-S_T,0) - max(K1-S_T,0)  =  K2 - K1
```

S_T terms cancel -> fixed payoff = a zero-coupon bond. No-arbitrage cost today:

`Box value = (K2 - K1) * e^(-rT)`

- Long box (pay now, collect K2-K1 later)
  = LENDING at the implied rate.
- Short box (collect now, pay K2-K1 later)
  = BORROWING.

Keep it risk-free: European cash-settled index options (SPX/XSP) - American can be assigned early and break the box. Hold to expiry.

Worked examples

1. Long call - K=100, c=4:

```
S_T = 90  -> profit  -4  (max loss = premium)
S_T = 104 -> profit   0  (breakeven = K + c)
S_T = 115 -> profit +11
```

2. Box as a loan - K1=5000, K2=5100, T = 0.5 yr, r = 5%:

```
payoff   = 100  (= $10,000/box x100)
fair cost= 100*e^(-0.025) = 97.53 (~$9,753)
buy @97.0 -> lend, get 100 -> ~6.3% > 5% (arb)
sell@98.0 -> borrow @ ~4.1% < 5% (cheap loan)
```